Unit 4, Lesson 1: Deriving Area Formulas for Trapezoids, Parallelograms, and Rhombi

Benchmark

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.

Learning Target

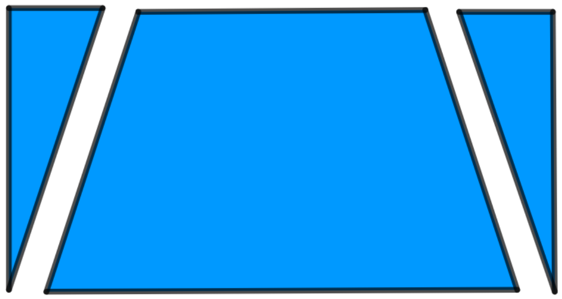
- ☐ I can use the formulas for the area of triangles and rectangles to derive formulas for the area of trapezoids, parallelograms, and rhombi.

4.1.1 Warm-Up: Notice and Wonder

Two figures are shown.

Figure 1

Figure 2



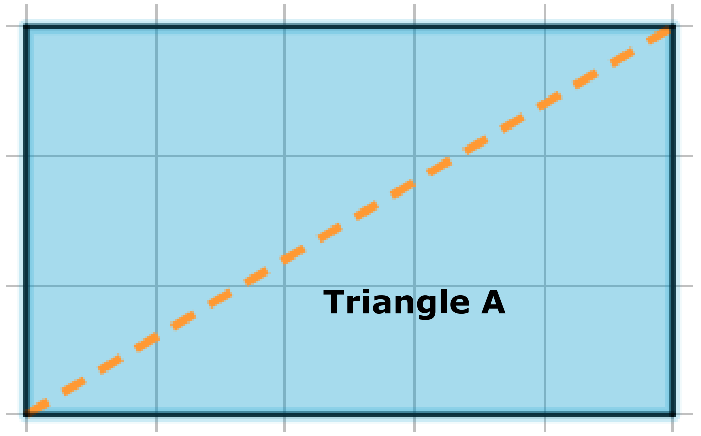
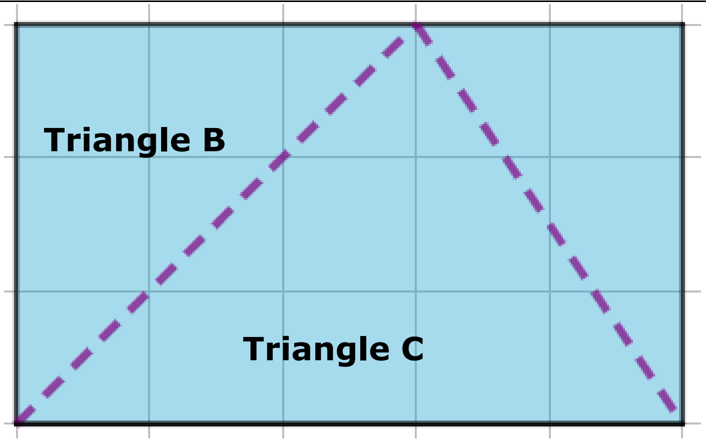
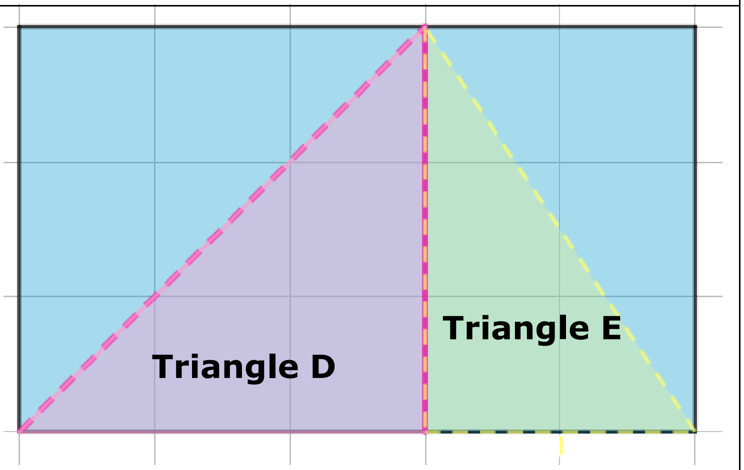
1. What do you notice about these two figures?

2. What do you wonder about the figures?

4.1.2 Refresher: Decomposing to Find the Area of a Triangle

Figures can be decomposed into smaller pieces that are then rearranged and composed into other shapes. Decomposing is useful when finding the area of a figure.

1. Complete the table.

The rectangle is divided into ______ triangles. The area of Triangle A is _____ of the area of the rectangle.	
The rectangle is divided into ______ triangles. The area of Triangle B is _____ of the area of the rectangle. The area of Triangle C is _____ of the area of the rectangle.	
The rectangle is divided into ______ triangles. The area of Triangle D is _____ of the area of the rectangle. The area of Triangle E is _____ of the area of the rectangle.	

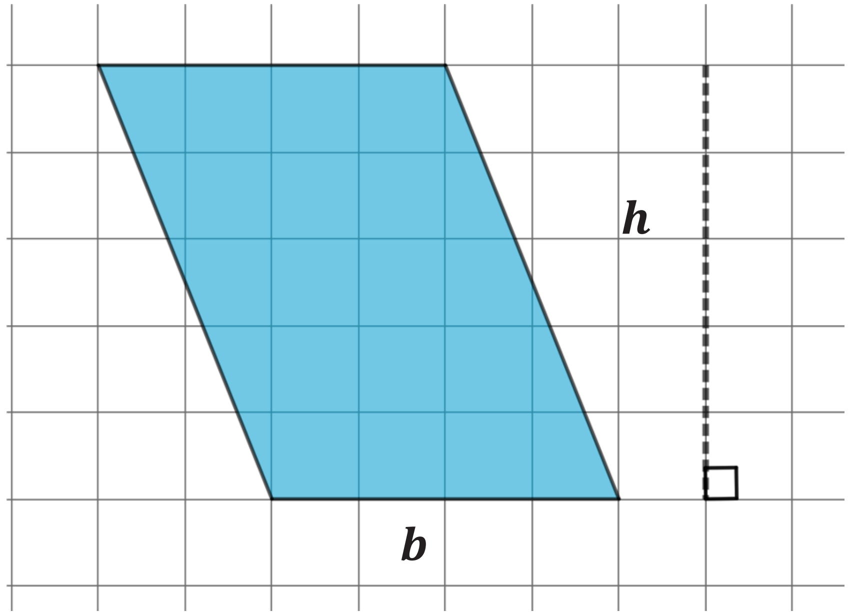
4.1.3 Activity: Deriving the Formula for the Area of a Parallelogram

A ***quadrilateral*** is a polygon with exactly four sides and four vertices.

A parallelogram is a type of quadrilateral. Not only does a parallelogram have four sides and four vertices, but its opposite sides are also parallel.

A ***parallelogram*** is a quadrilateral containing two pairs of parallel sides.

1. A parallelogram is shown on a grid with each gridline representing one unit.

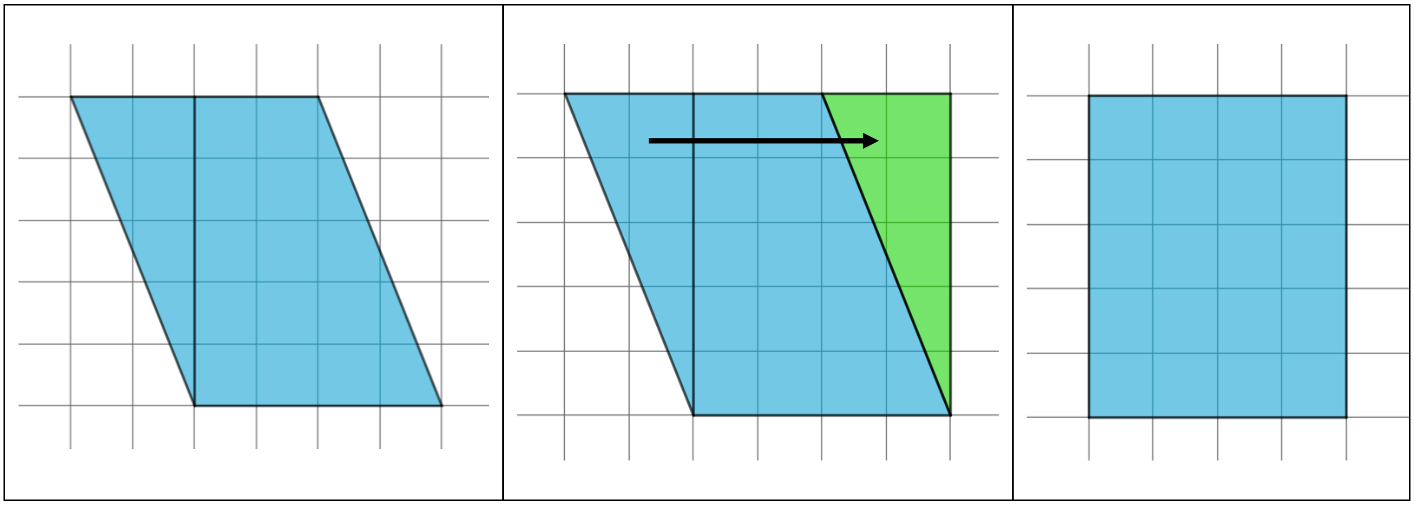


Complete each statement.

a. The base of the parallelogram is ______ units.

b. The height of the parallelogram is ______ units.

2. The images below show the same parallelogram being decomposed and then rearranged to form a rectangle.



Complete each statement.

a. The base of the rectangle is ______ units.

b. The height of the rectangle is ______ units.

3. Discuss with your partner the relationship between the area of the parallelogram and the area of the rectangle. Write a summary of your discussion.

4. Complete the statements.

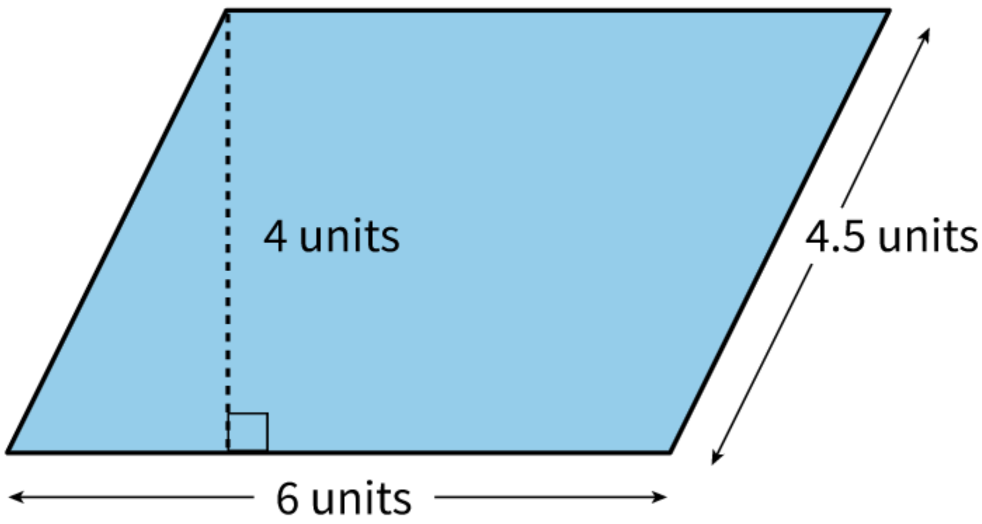
A parallelogram can be decomposed and rearranged into a
- ○ triangle
- ○ rectangle

with the same base and height.

The formula used to find the area of a rectangle is
- ○ $A = bh$.
- ○ $A = \frac{1}{2}bh$.

Therefore, the area of the parallelogram is
- ○ $A = bh$.
- ○ $A = \frac{1}{2}bh$.

5. A parallelogram with its dimensions labeled is shown.



a. Fill in the blanks to find the area of the parallelogram.

Area of parallelogram = $b \cdot h$

Area = _____ · _____

Area = _______ units^2

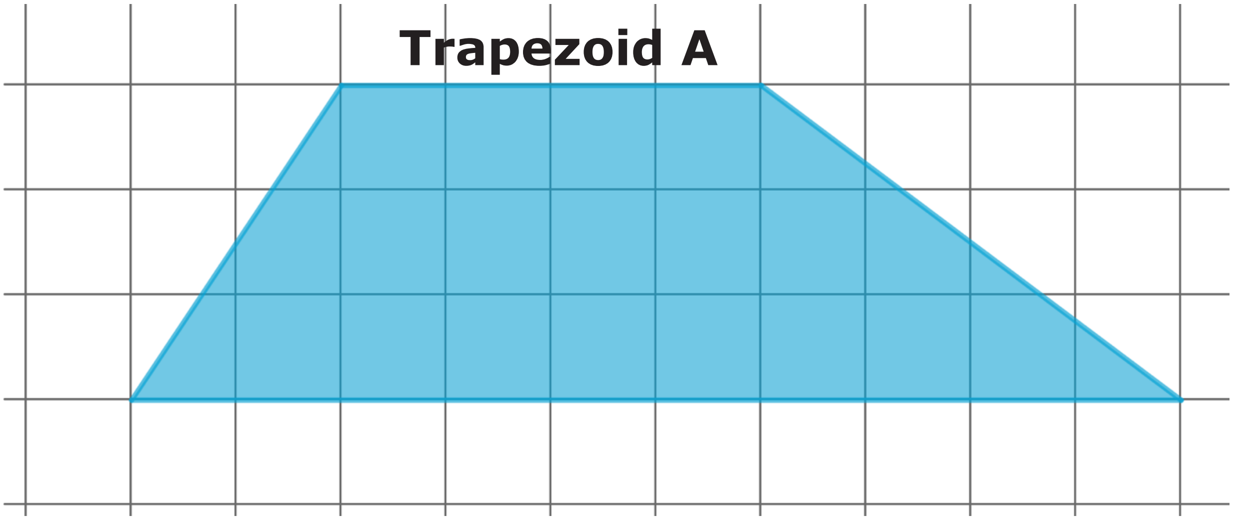
b. Three lengths are shown in the figure. With your partner, discuss the length that was not used to find the area of the parallelogram. Then, write an explanation for why this length was not used.

4.1.4 Activity: Deriving the Formula for the Area of a Trapezoid

A trapezoid is another type of quadrilateral.

A ***trapezoid*** is a quadrilateral with at least one pair of parallel sides.

Trapezoid A is shown on a grid with each gridline representing one unit.



The two parallel sides of a trapezoid are both called bases. To distinguish between the two bases, one is named base one, b_1, and the other base two, b_2.

1. On Trapezoid A, label one of the bases b_1 and the other b_2.

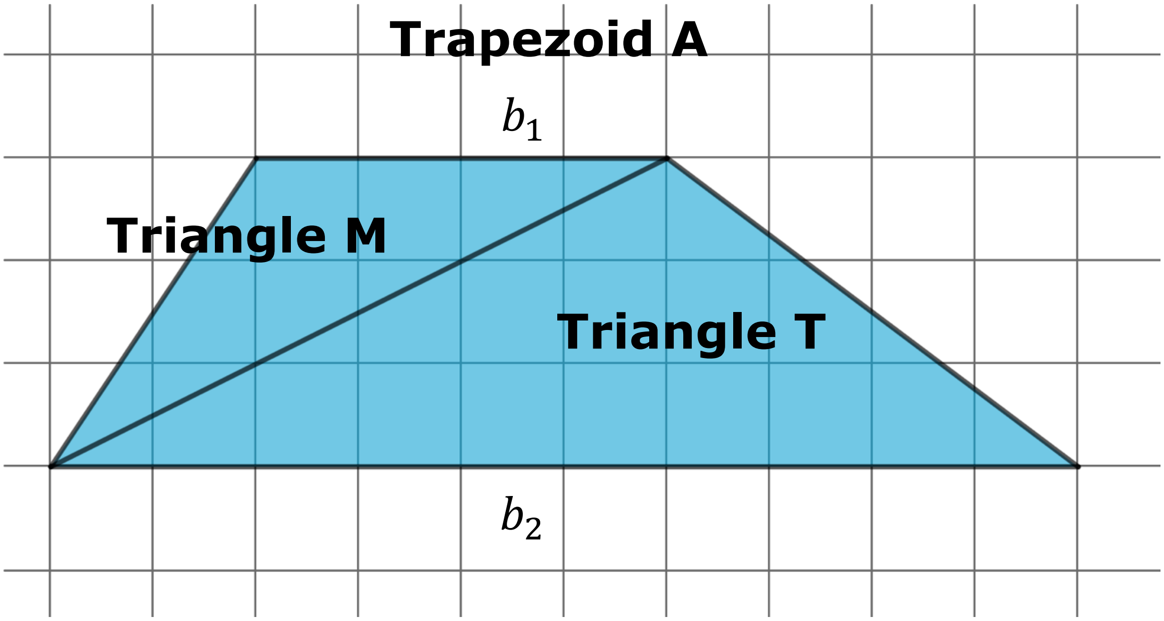
2. Complete each statement.

   a. The length of b_1 of Trapezoid A is ______ units.

   b. The length of b_2 of Trapezoid A is ______ units.

   c. The height of Trapezoid A is ______ units.

3. Trapezoid A is decomposed into two triangles and labeled, as shown below.



a. Complete the statements.

The base of Triangle M, b_1, is ______ units, and its corresponding height is ______ units.
The base of Triangle T, b_2, is ______ units, and its corresponding height is ______ units. When a trapezoid is decomposed into two triangles by drawing a diagonal, the height of both triangles will always be the same.

b. Davita wrote a formula for the area of Trapezoid A.

Davita's Equation
Area Trapezoid A = Area Triangle M + Area Triangle T $A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$

Explain how Davita determined this formula.

c. The formula Davita wrote can be used to find the area of any trapezoid. Complete the steps described in the table to rewrite the formula.

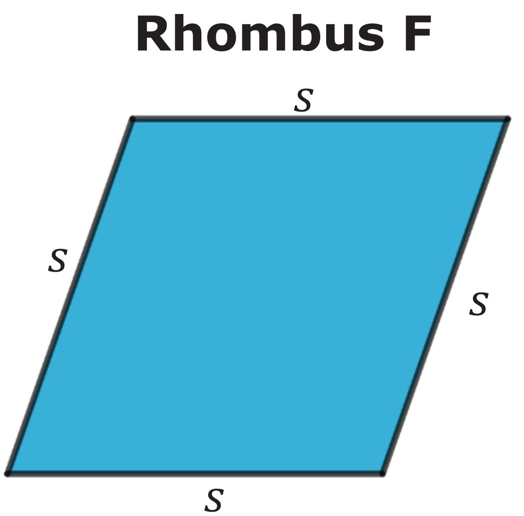
Area of Trapezoid A	
Davita's formula	$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$
Use the distributive property to rewrite with a common factor h	$A = \_\_\_\_\left(\frac{1}{2}b_1 + \frac{1}{2}b_2\right)$
Use the distributive property to rewrite with a common factor $\frac{1}{2}$	$A = \_\_\_\_h(b_1 + b_2)$

4.1.5 Activity: Deriving the Formula for the Area of a Rhombus

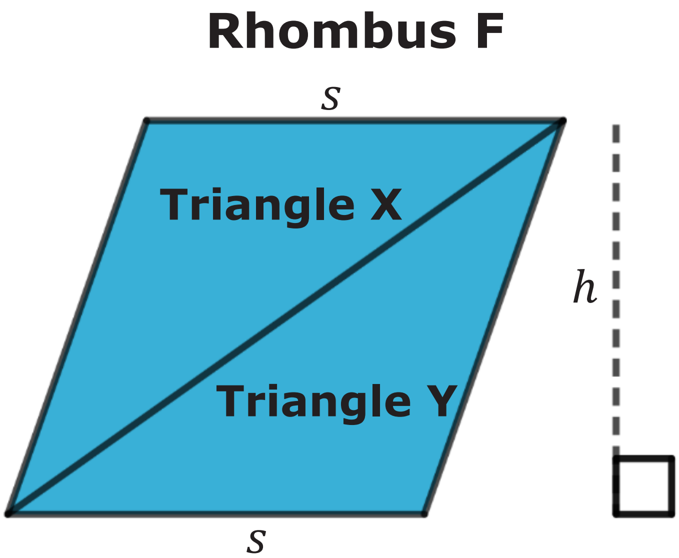
A rhombus is another type of quadrilateral.

A ***rhombus*** is a quadrilateral containing four sides of equal length.

Rhombus F is shown with four congruent sides.



Rhombus F is decomposed into two triangles, Triangle X and Triangle Y, as shown.



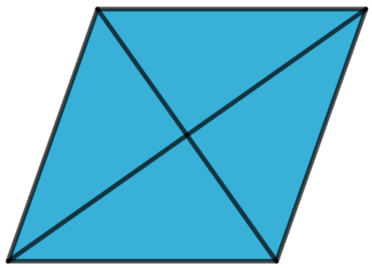
1. Felicia wrote expressions to represent the areas of the two triangles.

Area of Triangle X	Area of Triangle Y
$\frac{1}{2} \cdot s \cdot h$	$\frac{1}{2} \cdot s \cdot h$

Discuss with your partner whether you agree with the expressions Felicia wrote. Write a summary of your discussion.

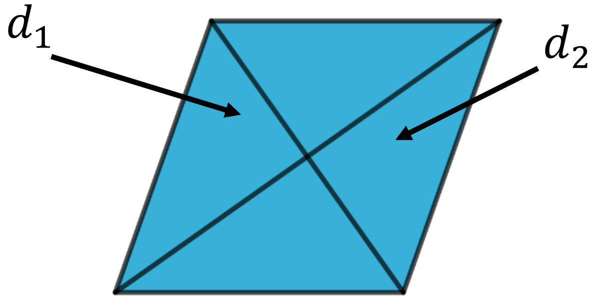
The diagonals of a rhombus can also be used to find its area.

The two diagonals of a rhombus are perpendicular to each other and divide each other in half.



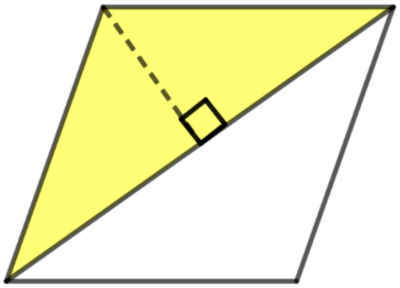
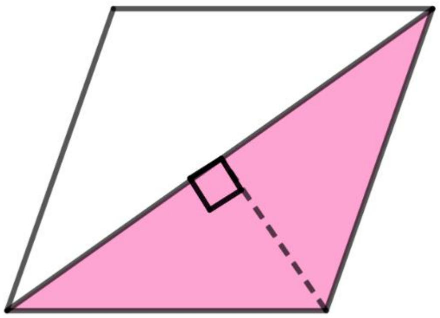
The two diagonals, d_1 and d_2, of Rhombus F are drawn and labeled below.

Rhombus F



Mark on the figure the two lengths that are equal to $\frac{1}{2}d_1$ and the two lengths that are equal to $\frac{1}{2}d_2$.

2. The first column of the table shows Rhombus F with parts highlighted. Use the lengths marked on Rhombus F in the previous question to fill in the blanks to find the area of the triangles.

Triangle	Area
Rhombus F 	Area $= \frac{1}{2}(b)(h)$ Area $= \frac{1}{2}(\_\_\_\_\_)(\_\_\_\_\_)$ Rewrite the area by multiplying any fractions. Area = ________________
Rhombus F 	Area $= \frac{1}{2}(b)(h)$ Area $= \frac{1}{2}(\_\_\_\_\_)(\_\_\_\_\_)$ Rewrite the area by multiplying any fractions. Area = ________________

3. Compare your work with your partner and resolve any differences, if necessary.

4. Work with your partner to express the area of Rhombus F as the sum of the area of the yellow triangle and the pink triangle found in the table.

Area of Rhombus F = ______________________________

5. Use words, values, or expressions from the bank shown to complete the statements. Items from the bank may be used more than once.

$\frac{1}{2}$	sum	$\frac{1}{4}(d_1)(d_2)$	diagonal

Drawing a _____________ of a rhombus divides the rhombus into two identical triangles. The area of the rhombus is equal to the _____________ of the areas of these two triangles. If one of these triangles is equal to _____________, then two of these triangles equal

$2 \cdot$ _____________. Therefore, the formula for the area of a rhombus using its diagonals can be written as

$A =$ ____$(d_1)(d_2)$.

4.1.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point for you in today's lesson on deriving the formula of parallelograms, trapezoids, and rhombi?